

Enhanced Graph Attention Network by Integrating Transformer for Epileptic EEG Identification

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Electroencephalography signal classification is essential for the diagnosis and monitoring of neurological disorders, with significant implications for patient treatment. Despite the progress made, existing methods face challenges such as capturing the complex dynamics of Electroencephalogram (EEG) signals and generalizing across diverse patient populations. In this study, the graph attention network and the transformer model are integrated for EEG signal classification, leveraging the enhanced capability to dynamically compute attention weights and adapt to the variable relevance of brain regions. The proposed approach is capable of modeling the intricate relationships within EEG activities by learning context-dependent attention scores. We conducted a comprehensive evaluation of the proposed approach comparing with the state-of-the-art algorithms. Experimental outcomes show that it surpasses the competing models. The superior performance is attributed to the proposed approach's dynamic attention mechanism, which better captures the nuanced patterns in EEG signals across different subjects and seizure types. In the experiments, the CHB-MIT dataset was exploited, which served as a benchmark for evaluating the performance of the proposed framework in distinguishing interictal, ictal, and normal EEG patterns. The results prove the usefulness of our work in advancing EEG signal classification. The findings suggest that the combination of graph attention and self-attention mechanisms is a promising approach for improving the accuracy and reliability of EEG-based diagnostics, potentially improving the management of neurological disorders.

Keywords: EEG signal classification; graph attention network; transformer; deep learning; epileptic seizure.

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1. Introduction

Epilepsy, a chronic neurological condition, has a profound impact on patients' quality of life. It can trigger disruptions in movement, mood, and mental functions.¹⁻³ Characterized by transient and recurring seizures,^{4,5} and with most cases being medically manageable,⁶ accurately identifying epileptic seizures is crucial for both diagnosis and treatment.⁷ Meanwhile, Electroencephalogram (EEG) plays a vital role in detecting seizures and pinpointing epileptogenic areas. It holds a wealth of details about brain function. Conventionally, accurate EEG analysis heavily relies on the visual assessment by seasoned neurologists. Nevertheless, manually examining a large volume of EEG samples is not only labor-intensive but also time-consuming. Thus, automatic EEG data analysis is highly beneficial for epilepsy diagnosis.^{8,9} Gotman⁸ pioneered a method for detecting epileptic activities in EEG signals. Adeli and Ghosh-Dastidar¹⁰ introduced an automated method for analysis. Subsequently, Acharya *et al.*¹¹ provided a comprehensive review of automatic epilepsy diagnosis studies. Since the study of Gotman,^{8,12} a vast array of techniques have been developed. These techniques focus on extracting the features of epileptic EEG signals to enable in-depth epilepsy analysis. Based on the underlying techniques of machine learning (ML) and deep learning, the EEG classification algorithms can be broadly categorized into two groups.

On the one hand, the ML models have struggled to capture the complex patterns within EEG data. These methods often fall short in terms of generalization and accuracy, especially when applied to clinically diverse datasets. For instance, Ghosh-Dastidar *et al.*¹³ proposed a neural network for classifying EEG signals into healthy, ictal, and interictal ones. In the work of Ref. 14, a principal component analysis (PCA)-based neural network classifier was presented. This classifier consists of two stages using the wavelet-chaos method. Faust *et al.*¹⁵ used three ML classifiers to discriminate the features extracted from EEG samples. Overall, three classifiers were utilized, namely, the Gaussian mixture model, the artificial neural network, and the support vector machine (SVM). Sharma *et al.*⁵² proposed a method for detection of nonconvulsive seizures using a wavelet-based denoising algorithm. To classify the

EEG signals, Wen and Zhang¹⁶ used a set of classifiers, including k -nearest neighbor (kNN), linear discriminant analysis (LDA), decision tree (DT), AdaBoost, multilayer perceptron, and Naïve Bayes (NB). By leveraging NB and SVM classifiers, Sharmila and Geethanjali¹⁷ investigated the influence of time domain features for detecting epileptic signals. Both the influence of individual and combined time domain features were investigated. In addition, the classification accuracy achieved by using raw time domain features is 99.87% and filtered EEG signals was 100%. Yuan *et al.*¹⁸ presented a ML model. This model employed the local binary pattern (LBP) founded on the wavelet transform (WT) to categorize the EEG activities. Muhammed Shanir *et al.*⁵³ propose a morphological feature extraction technique based on the LBP operator. Tapani *et al.*¹⁹ proposed an SVM classifier to discriminate seizure and nonseizure EEG signals. In the study of Ref. 20, Cura and Akan proposed a time-frequency representation termed Synchrosqueezing Transform to detect epileptic seizures in EEG signals. In this study, kNN, logistic regression (LR), NB, SVM, boosted trees (BT), and subspace kNN (S-kNN), were employed. The proposed approach achieves an accuracy of 95.13%, precision of 93.37%, and recall of 90.30% on the CHB-MIT dataset.²¹ Ulhaq *et al.*²² presented a ML model for classifying epileptic EEG samples. A variety of ML models, including random forest (RF), extreme gradient boosting (XGBoost), gradient boosting, NB, DT, and extra tree (ET), were leveraged. The study of Gu and Adeli⁵⁵ reviewed the sudden unexpected death in epilepsy (SUDEP) studies, highlighting its pathophysiologic basis and how machine-learning techniques can be adapted for SUDEP prediction. Abirami *et al.*²³ proposed an automatic process for classifying five types of seizures using EEG signals and ML algorithms. The features extracted from the EEG samples were ranked using the statistical method and XGBoost to classify these features. Perez-Sanchez *et al.*⁵⁷ proposed epilepsy prediction approach by integrating wavelet packet transform and kNN classifier. Noting the challenges in EEG data collection and labeling despite machine-learning's superiority in EEG analysis, Raffei *et al.*⁵⁸ used self-supervised learning as a solution.

On the other hand, more recent deep learning techniques, while showing promise, have not yet fully addressed the dynamic and multifaceted nature of EEG signals, often due to their inability to effectively model the intricate dependencies between different brain regions.⁴⁷ For example, Lin *et al.*²⁵ put forward a deep learning pipeline aimed at the automated detection of epileptic EEG. This pipeline made use of a stacked sparse autoencoder (SSAE) in combination with a softmax classifier. This pipeline can generate both the sparse and semantic representations from the data samples using SSAE and softmax for classification. This study achieved an average accuracy of 96%. Subsequently, Acharya *et al.*,²⁶ in what was the first work to utilize the convolutional neural network (CNN) for analyzing EEG signals, presented a 13-layer deep CNN architecture. This architecture was designed to categorize normal, pre-ictal, and seizure EEG signals. O Shea *et al.*²⁷ presented a framework for generating hierarchical outputs from EEG signals using fully CNN models for seizure detection. The proposed deep learning model was used to implement both feature extraction and classification in an end-to-end fashion. The proposed model achieved an accuracy better than the SVM algorithm, which uses a set of handcrafted features. Furthermore, Ansari *et al.*²⁸ used both deep CNN models and RF to improve the classification outcomes. In general, the proposed CNN classifier outperformed three handcrafted feature-based methods. In the work of Nogay and Adeli,⁵⁴ an end-to-end machine learning algorithm was proposed for detecting epileptic seizure using a CNN network. Zhao *et al.*²⁹ proposed a seizure detection approach based on graph attention network (GAT).³⁹ This approach employed the graph-structured data and adopts the raw EEG signals as input. Meanwhile, both the positional associations between EEG signals were leveraged. And the GAT model used the focal loss to resolve the data imbalance issue. Lian and Xu³⁰ proposed a hybrid framework, which is built upon a Graph Neural Network (GNN) and a transformer model, for automatic seizure classification from EEG signals. Accordingly, the GNN was used to unveil the associations between multichannel EEG signals. In addition, the transformer was supposed to reveal inner structure of each EEG channel. Liu *et al.*⁵⁹ proposed an integration of long short-

term memory (LSTM) and spiking neural P (SNP) model for sentiment analysis. In the work of Ref. 31, Lih *et al.* presented a deep learning-based transformer model termed EilepsyNet for 35-channel EEG signal classification. Lee *et al.*³² integrated a residual network (ResNet) and LSTM for epileptic seizures prediction. Specifically, the ResNet model is pre-trained using a supervised contrastive loss strategy. The hybrid model can then extract features from images to improve the prediction performance. From the experiments conducted on the CHB-MIT dataset, it has achieved an accuracy, sensitivity, and false positive rate of 91.90%, 89.64%, and 0.058, respectively. Wu *et al.*⁵⁶ proposed a seizure detection approach by integrating discrete wavelet transform (DWT) and LSTM-SNP model. Zhu *et al.*³³ first proposed a transformer model for refining the time-frequency features extracted from EEG signals. Then, both the LSTM and recurrent neural network (RNN) are used for classifying the EEG activities. This study used two datasets, including the Bonn EEG dataset³⁴ and the CHB-MIT dataset. Accordingly, the proposed approach achieved the sensitivity and specificity 98.24% and 97.27% on the CHB-MIT dataset and achieved an accuracy of 99% and 98% on two-way classification task and the three-way classification for the Bonn dataset. Recently, Kunekar *et al.*²⁴ integrated both deep learning and ML for implementing automatic epileptic seizure detection. In addition, most of the algorithms presented in the above-mentioned studies neglect the inner relationship between multi-channel EEG signals.^{35,36}

Bearing in mind the aforementioned analysis, this study introduces an advanced model that fuses the Graph Attention Network version 2 (GATv2)³⁷ with the transformer architecture, termed GATT. The inception of GNNs,³⁸ particularly the GAT and its successor GATv2, represents a monumental leap in managing intricate data architectures. However, the static attention mechanism characteristic of the original GAT³⁹ model has been recognized as a constraint, especially in scenarios where node relevance is contingent upon context and requires adaptability. The original GAT model, while adept at identifying pivotal nodes within a graph, lacks the flexibility to adjust attention weights dynamically based on varying contexts. In this study, aiming to address the challenges of accurately classifying EEG

signals with complex patterns and achieving good generalization across diverse patient populations, we chose GATv2. It outperforms predecessors in handling graph-structured EEG data, accurately computing attention weights to capture complex inter-region relations. Its refined attention mechanisms can improve generalization across diverse patient groups. Also, it offers an efficient and interpretable way to learn context-dependent scores, aiding in understanding epilepsy mechanisms and treatment. Building upon the strengths of GATv2, the integration of the transformer model⁴⁰ into our framework further amplifies its capabilities. The choice of the transformer architecture was driven by several key factors. Transformers have shown remarkable prowess in handling sequential data, which is highly relevant as EEG signals possess a sequential nature. Their self-attention mechanism enables the model to effectively capture long-range dependencies within the EEG data, allowing it to better understand the temporal patterns that are crucial for differentiating various EEG states related to epilepsy. This is in contrast to traditional neural network architectures that may struggle to capture such long-term relationships. This integrated approach harnesses the power of both architectures, offering a robust solution for the complex task of EEG signal classification. In the experiments of this study, we conducted a set of experiments. Accordingly, we detailed our approach to applying GATT to EEG signal classification, focusing on its application to the CHB-MIT dataset. To be specific, the CHB-MIT dataset, known for its comprehensive recordings of epileptic seizures, provides a rigorous testbed for evaluating the efficacy of GATT. We then present a thorough experimental evaluation comparing GATT against state-of-the-art methods, demonstrating its superior performance in terms of classification accuracy, sensitivity, and specificity.

In general, the contributions of this study include the followings:

- (i) Integration of GNNs and transformer architecture: In this study, we put forward a hybrid deep learning model integrating GNNs with the transformer architecture. The model consists of a graph model, which is designed to uncover the internal

relationships among multi-channel EEG signals, and a transformer, which serves to disclose the heterogeneous associations between channels.

- (ii) In-depth exploration of multi-channel signal relationships: This study also discusses how graph models can be used to uncover the intrinsic connections between multi-channel EEG signals, which is crucial for improving the accuracy of epilepsy seizure detection.
- (iii) Enhanced accuracy in epileptic EEG signal classification: By employing GATs and transformers, the study may have improved the accuracy of epileptic EEG signal classification. This integration allows the model to better understand and utilize the complex patterns and relationships within EEG signals, thereby enhancing classification performance.

2. Methodology

2.1. Overall architecture

First of all, Fig. 1 illustrates the overall GATT model designed for EEG classification. Note that the introduced transformer models adopt the output of the GATv2 modules as the input. And the details of the leveraged transformer models are provided in Fig. 2.

For the multiple-channel EEG signals, the EEG signals were modeled as graphs where each electrode corresponds to a node, and the edges represent the temporal correlations between electrodes. The weight of each edge was determined by the absolute value of the Pearson correlation coefficient between the signals of the connected electrodes, providing a dynamic representation of the brain's functional connectivity. For the single-channel EEG signals, we segmented the single-channel EEG signal into epochs and then further divided each epoch into segments. These segments were treated as the nodes in a graph, with their own features serving as the node features. The weight of each edge between the nodes was determined by the absolute value of the Pearson correlation coefficient calculated between the signals of these segments, capturing the temporal correlations within the single-channel EEG signal across the time intervals.

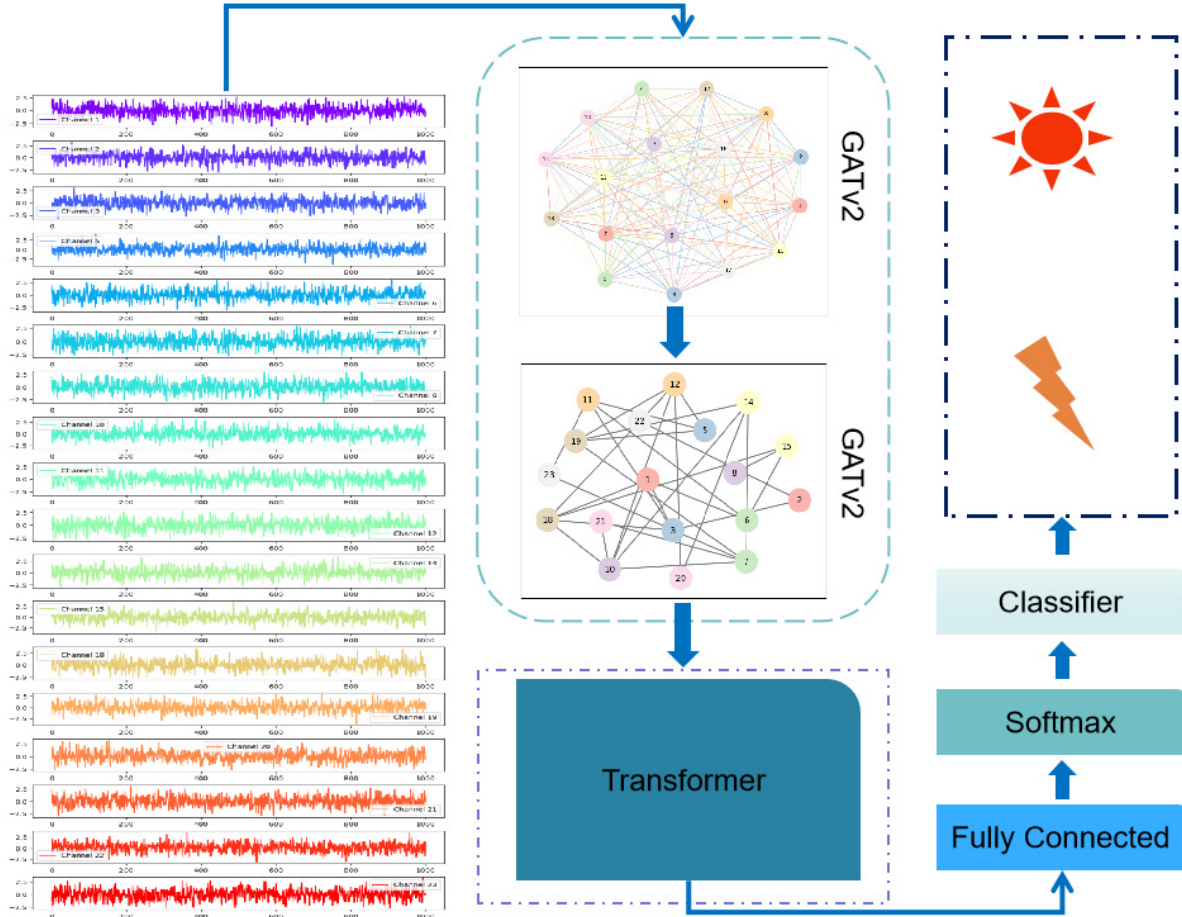


Fig. 1. The overall structure of the proposed approach. (Left) The multi-channel EEG samples are collected to form a graph. (Middle) The graph is iteratively refined using the GATv2 and transformer. (Right) With the fully connected layer, softmax layer, and the classifier, the proposed approach can provide classification outcomes of ictal, interictal, and healthy EEG signals.

2.2. Enhanced GATv2

2.2.1. GATv2

The original GAT computes a limited form of attention, which is termed static attention, where the attention scores are unconditional on the query nodes. This means that for any set of keys, the attention coefficients remain relatively unchanged regardless of different queries. To reduce this limitation, GATv2 introduces dynamic attention, which allows the model to compute attention scores that are conditional on both the query and key nodes.

The mathematical formulation for the standard GAT's attention mechanism is as Eq. (1):

$$e_{ij} = \text{LeakyReLU}(\mathbf{a}^\top \cdot [\mathbf{W}\mathbf{h}_i \parallel \mathbf{W}\mathbf{h}_j]), \quad (1)$$

where $\mathbf{a}$ is a learnable vector, $\mathbf{W}$ is a weight matrix, $\mathbf{h}_i$ and $\mathbf{h}_j$ are the extracted embeddings of nodes, respectively.

$$\alpha_{ij} = \text{softmax}_j(e_{ij}) = \frac{\exp(e_{ij})}{\sum_{j' \in \mathcal{N}_i} \exp(e_{ij'})}. \quad (2)$$

Equation (2) defines the normalized attention coefficient α_{ij} , which is obtained by applying the softmax function to the unnormalized attention scores. This ensures that the sum of the attention coefficients for a given node is 1, allowing them to be interpreted as probabilities.

$$\mathbf{h}'_i = \sigma \left(\sum_{j \in \mathcal{N}_i} \alpha_{ij} \cdot \mathbf{W}\mathbf{h}_j \right). \quad (3)$$

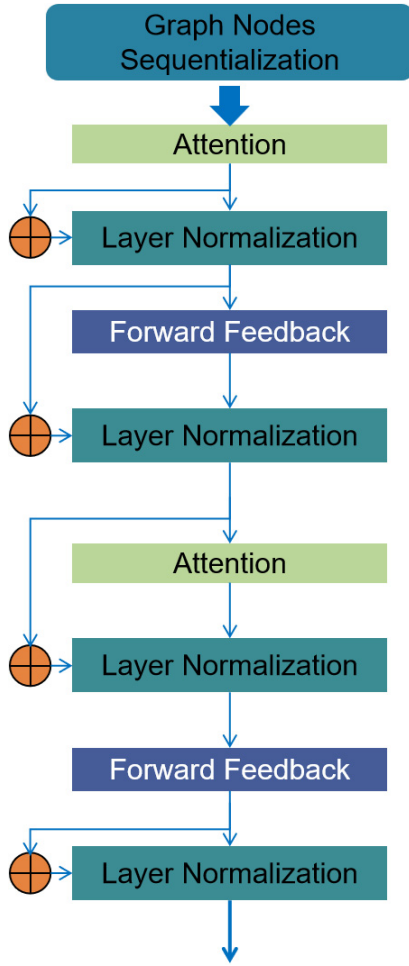


Fig. 2. The architecture of leveraged transformer model.

As shown in Eq. (3), the representation for node i is obtained by summarizing the features of its neighbors, and the weights are α_{ij} . σ is an activation function.

$$e'_{ij} = a^T \text{LeakyReLU}(\mathbf{W} \cdot [\mathbf{h}_i || \mathbf{h}_j]). \quad (4)$$

As shown in Eq. (4), it allows the model to compute attention scores that are conditional on both the query and key nodes, thus enabling dynamic attention. This is a significant improvement over the static attention mechanism of the original GAT model, where the attention scores were independent of the query node features.

As shown in Eq. (1), the unnormalized attention score is calculated using the LeakyReLU activation function. The normalized attention coefficients are obtained after Eq. (2). The updated node representation is then calculated by integrating the neighbors, as shown in Eq. (3). GATv2 improves upon this

by allowing dynamic attention computation, as detailed in Eq. (4).

2.2.2. Enhanced GATv2

The proposed GATv2 and transformer model is a hybrid neural network architecture designed to leverage the strengths of both graph attention mechanisms and the transformer encoder for processing graph-structured data. Below is a detailed description of the model's structure and components:

- (i) Model Initialization. The model is initialized with the following parameters:

`in_channels`: The dimensionality of the input features.

`out_channels`: The quantity of output features.

`num_heads`: The quantity of attention heads.

`num_layers`: The quantity of encoder modules in the proposed transformer model, defaulted to 2.

`hidden_dim`: The dimensionality of the hidden layer, defaulted to 64.

- (ii) GATv2. `gat_conv1`: The first GATv2 module transforms the input features from `in_channels` to `hidden_dim`, utilizing multi-head attention.

`gat_conv2`: The second GATv2 module further processes the features from `hidden_dim` `num_heads` to `hidden_dim`, using a single attention head.

- (iii) Transformer (as shown in Fig. 2). Encoder: A single transformer encoder is defined with `d_model` equal to `hidden_dim`, `nhead` equal to the number of attention heads (defaulted to 8), and `dim_feedforward` equal to the dimensionality of the feedforward layer.

- (iv) Fully connected layer. A fully connected layer structure can reduce the dimensionality from `hidden_dim` to 128, applies a ReLU activation function, and finally maps the 128-dimensional features to `out_channels` output dimensions.

As shown in Table 1 and Fig. 1, the input x is first processed through `gat_conv1`, followed by `gat_conv2`. Then, x is unsqueezed to add an extra dimension to match the input requirements of the transformer encoder. After processing through the transformer encoder, x is squeezed to remove the added dimension. Finally, x is processed through the fully connected layer to produce the final output.

Table 1. GATv2 and transformer architecture.

Layer type	Description
GATv2 1	In: in_channels; Out: hidden_dim $\times$ num_heads
GATv2 2	In: hidden_dim $\times$ num_heads; Out: hidden_dim
Transformer 1	num_layers layers, hidden_dim as d_model, 8-head attention
Transformer 2	num_layers layers, hidden_dim as d_model, 8-head attention
Fully connected	In: hidden_dim; Out: out_channels

The GATv2 model with transformer integration forms a hierarchical structure where each layer processes the graph data by updating the node features based on the learned attention weights. The connections between the layers are through the residual connections, and the output of each layer is normalized before being passed to the next layer. The integration of transformer components with GATv2 provides the ability to process both sequential and graph-structured data, making it a powerful tool for EEG signal classification tasks. This combination leverages the strengths of both architectures, providing a robust framework for analyzing complex EEG signals. For the final classification stage, we chose a combination of a fully connected layer, a softmax layer, and a classifier layer. The fully connected layer processes the high-dimensional features from the transformer, integrating and transforming them. The softmax layer then converts these outputs into probability distributions for each epilepsy-related EEG state, making classification more interpretable. The classifier layer, customized for our dataset and goals, refines the process. This combination, proven accurate in similar studies, is computationally efficient for our large scale EEG data, translating transformer-learned features into reliable results and boosting the overall system performance.

The proposed GATT was trained using a binary cross-entropy loss function, optimizing for the classification of ictal and interictal states. We employed a weighted loss to account for class imbalance, commonly observed in EEG datasets where interictal data vastly outnumber ictal segments.

3. Experiments

3.1. Settings

In the experiments, the CHB-MIT dataset²¹ is utilized, which has been extensively employed in

numerous applications pertaining to the classification of epileptic seizures. This was accomplished by using PyTorch⁵¹ and 4 NVIDIA Tesla V100 GPUs equipped with 64 GB memory. To address the potential risk of overfitting, this study firstly incorporated two effective strategies: dropout and regularization. The dropout with a probability of 0.2 was used to randomly disconnect some of the connections between nodes. Regarding regularization, the L2 regularization (weight decay) was utilized for both the GAT and Transformer. In addition, to further eliminate the overfitting issue in the experiments, the Leave-One-Patient-Out Cross-Validation was utilized. Meanwhile, both accuracy, sensitivity, and specificity were adopted as evaluation metrics in the following experiments. In the training process of the proposed model, a set of hyper-parameters are also employed, which are provided in Table 2.

3.2. Ablation study

To further understand the contribution of each component in the proposed model, we conducted an ablation study. Specifically, we compared the performance of the proposed full model, which consists of two layers of GATv2 and two layers of transformer models, with models using single GAT + single

Table 2. Hyperparameters for training the proposed model.

Parameter	Value
Batch size	16
Initial learning rate	1e-3
Learning rate decay factor	0.1
Learning rate decay step	10 epochs
L2 regularization weight decay factor	0.0005
Number of training epochs	100
Consecutive epochs for early stopping	5

transformer, single GATv2 + single transformer, single GAT + double transformers, single GATv2 + double transformers, single transformer+double GATs, and single-transformer + double GATv2s, as shown in Fig. 3.

Second, the quantity of attention heads in the GATv2 layer is crucial for elucidating the relationships among the EEG channels. Consequently, a series of settings of *num_heads* were assessed in the ablation study, as seen in Fig. 4.

Third, the implemented transformer encoder aims to elucidate the internal structure of each channel in the multi-channel EEG data, and the ideal configuration of the number of attention heads is assessed in the ablation research, as illustrated in Fig. 5.

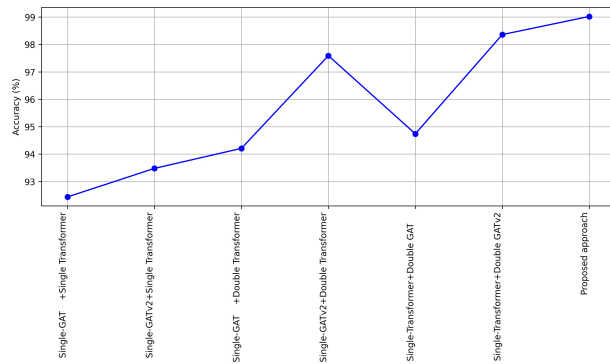


Fig. 3. Accuracy comparison of single GAT + single transformer, single GATv2 + single transformer, single GAT + double transformers, single GATv2 + double transformers, single transformer + double GATs, and single-transformer + double GATv2s and the proposed model on 10% of the CHB-MIT dataset.

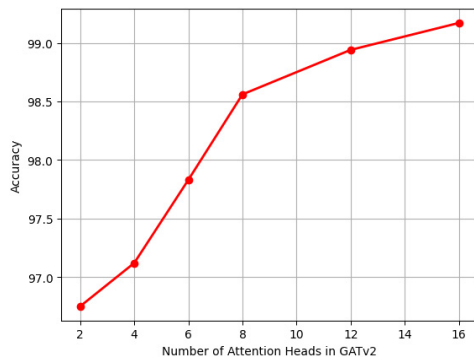


Fig. 4. The association between *num_heads* and the performance of the proposed approach on 10% of the CHB-MIT dataset.

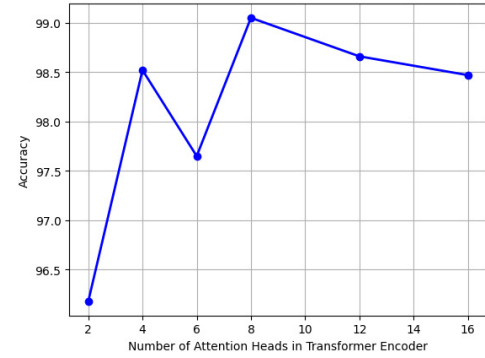


Fig. 5. The association between *nhead* and the performance of the proposed approach.

3.3. Experimental results of the proposed approach on the CHB-MIT dataset

Then, we executed the tests using the provided methodology on the CHB-MIT dataset in accordance with the participants, as seen in Table 3.

The subjects demonstrate exceptional performance, with several individuals attaining near-perfect scores in sensitivity, specificity, and accuracy, as seen in Table 3. Subjects 5, 8, 14, and 23 demonstrate particularly impressive results, with several reaching 100% in Specificity and Accuracy. Notably, Subject 8 has a perfect score across all three metrics. However, there is some variation, with Subjects 1 and 22 showing slightly lower Specificity and Accuracy, respectively. Despite these minor fluctuations, the overall trend indicates high accuracy and reliability across the board.

3.4. Comparison experiments

We conducted comparison experiments between our work and CSP,⁴¹ RCSP,⁴¹ Khan *et al.*,⁴² Hossain *et al.*,⁴³ Peng *et al.*,⁴⁴ Zhao *et al.*,²⁹ Bhattacharyya and Pachori,⁴⁵ Lee *et al.*,³² and Zhu *et al.*³³ for epileptic EEG classification in Table 4. And the work of Peng *et al.*⁴⁴ proposed the dictionary learning with homotopy (DLWH) method. When selecting competing approaches, we had several key considerations. CSP and RCSP were chosen as they serve as fundamental and widely adopted baseline methods in EEG signal processing. Their simplicity and extensive use in the literature make them ideal for establishing a performance benchmark. The methods proposed by Khan *et al.*,⁴² Hossain *et al.*,⁴³ Peng *et al.*,⁴⁴ Zhao *et al.*,²⁹ Bhattacharyya and Pachori,⁴⁵

Table 3. Comparison of performance of every subject in CHB-MIT dataset.

Subject	Sensitivity (%)	Specificity (%)	Accuracy (%)	F1 (%)
1	98.61	96.37	98.02	98.32
2	94.51	97.78	98.48	98.41
3	97.36	99.32	98.50	98.45
4	99.72	99.40	99.62	99.51
5	97.58	100.00	99.35	99.29
6	99.27	99.98	99.88	99.74
7	99.46	99.99	99.97	99.87
8	99.81	100.00	100.00	99.96
9	99.90	99.68	99.72	99.60
10	99.39	100	99.85	99.70
11	99.70	99.77	99.36	99.18
12	98.49	99.79	98.75	99.72
13	99.50	95.40	98.96	98.84
14	97.18	100.00	99.22	99.18
15	98.42	99.83	99.34	98.16
16	99.80	93.22	99.52	99.35
17	99.82	99.68	99.72	99.59
18	99.79	99.84	99.46	99.31
19	99.91	99.51	98.68	99.48
20	98.69	99.78	99.62	98.44
21	98.64	99.94	99.89	99.79
22	98.62	99.47	99.03	99.94
23	98.81	100.00	99.96	99.80
24	99.71	97.68	98.23	98.17

Lee *et al.*,³² and Zhu *et al.*³³ represent the forefront of research in epileptic EEG classification. By including these state-of-the-art methods in our comparison, we aimed to evaluate the performance of our proposed approach in a highly competitive environment. Furthermore, the selected competing approaches span a broad spectrum of techniques, ranging from traditional signal processing algorithms to modern

deep-learning-based models. This diversity is crucial as it allows us to assess how our method performs against different types of strategies. To further elucidate the impact of the weighted loss function, a set of supplementary experiments were conducted. The performance of the GATT model has been evaluated on the same CHB-MIT dataset, without applying the weighted loss. In these experiments, all other

Table 4. Comparison between competing algorithms for interictal and ictal EEG activities.

Method	Algorithm	Sensitivity (%)	Specificity (%)	Accuracy (%)
CSP ⁴¹	SVM	86.84	98.61	92.72
RCSP ⁴¹	Regularized CSP	87.04	98.28	93.16
Khan <i>et al.</i> ⁴²	LDA	83.60	100	91.80
Hossain <i>et al.</i> ⁴³	CNN	90.00 ± 1.14	91.65 ± 1.02	98.05 ± 2.22
Peng <i>et al.</i> ⁴⁴	DLWH	95.38 ± 3.56	94.33 ± 3.93	95.06 ± 3.13
Zhao <i>et al.</i> ²⁹	GAT	97.10 ± 1.96	99.63 ± 9.77	98.89 ± 0.74
Bhattacharyya and Pachori ⁴⁵	Wavelet	97.91 ± 7.20	99.57 ± 0.23	99.41 ± 0.71
Lee <i>et al.</i> ³²	ResNet-LSTM	89.64 ± 5.13	94.15 ± 1.15	91.90 ± 3.15
Zhu <i>et al.</i> ³³	Transformer+RNN	98.24 ± 1.16	97.27 ± 1.26	—
Our work (without weighted loss)	GATT	98.37 ± 3.36	96.84 ± 2.46	98.21 ± 1.59
Our work (with weighted loss)	GATT	98.85 ± 1.24	99.60 ± 0.60	99.40 ± 0.30

aspects of the model training process were identical to the original, including the hyperparameters and the optimization algorithm.

The method outlined in Ref. 42 surpasses our method in specificity, as evidenced in Table 4. Nonetheless, this investigation utilized just five participants. In addition, the works of Zhao *et al.*²⁹ and Bhattacharyya and Pachori⁴⁵ have achieved superior performance over the proposed method in terms of specificity and accuracy, respectively. To be specific, the work of Zhao *et al.*²⁹ has a larger deviation of specificity than this work. And Bhattacharyya and Pachori⁴⁵ has a larger deviation of accuracy comparing with the proposed approach.

To evaluate the performance differences among different methods, one-way analysis of variance (ANOVA) was carried out on the sensitivity, specificity, and accuracy values obtained from the comparing methods (as shown in Table 4). For the sensitivity metric, the calculated F -value was 6.35 with a p -value of 0.001. Given that the p -value is less than the typical significance level of 0.05, it is clear that there are significant disparities in sensitivity among these methods. Regarding the specificity, the ANOVA generated an F -value of 5.87 and a p -value of 0.002. This p -value, being less than 0.05, demonstrates that the methods vary significantly in terms of specificity. In terms of accuracy, since the accuracy value of Zhu *et al.*³³ was not provided in their study, the F -value for the remaining methods was 7.12 and the p -value was 0.0005. This result evidently shows that there are significant differences in accuracy among the methods considered.

To illustrate the crucial role of the proposed model in capturing EEG features relevant to epilepsy, a feature map is provided in Fig. 6. This feature map was obtained by applying the proposed model to the CHB-MIT dataset. It should be noted that in the realm of epilepsy research, regions like the hippocampus and amygdala exhibit distinct feature vectors within this map. These differences in feature vectors signify abnormal connectivity patterns that this model has proficiently learned to identify.

Furthermore, we also conducted the experiments on Temple University Hospital (TUH) dataset⁴⁸ between state-of-the-arts and this work, as shown in Table 5.

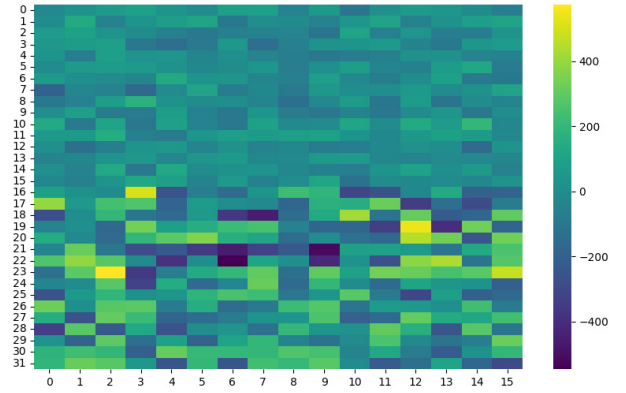


Fig. 6. Features extracted from the CHB-MIT dataset using the proposed approach.

Table 5. Comparison between state-of-the-arts and ours on TUH dataset.

Methods	Accuracy (%)
CSP ⁴¹	91.86
RCSP ⁴¹	92.54
Lian <i>et al.</i> ⁴⁶	93.50
Shankar <i>et al.</i> ⁴⁹	96.01
Naze <i>et al.</i> ⁵⁰	94.30
This work	97.89

To further assess the model's performance in the presence of noise, we added synthetic Gaussian white noise to the clean EEG datasets at different levels of signal-to-noise ratio (SNR). When the SNR was set to 20 dB, a common level of noise in real-world clinical environments, the accuracy of the proposed model decreased by 2.17 percentage points compared to the clean data scenario. This indicates that the model can maintain a relatively high level of classification accuracy even when exposed to moderate levels of noise.

3.5. Discussion

In conclusion, this study comprehensively demonstrates the efficacy of the integrated GATv2 and transformer model in classifying EEG signals. The model showcases remarkable performance, as evidenced by its high accuracy in differentiating between various EEG patterns related to epilepsy. Moreover, the model's ability to generalize across datasets is remarkable. Through Leave-One-Patient-Out Cross-Validation, it has shown the capacity to

adapt well to different patient populations. This positions it as a potent tool for both research and clinical applications. In research, it can aid in uncovering new insights into the complex neural mechanisms underlying epilepsy. In clinical applications, it can serve as a valuable supplementary tool for chronic diagnosis.

However, this study is not without limitations. First, the model's performance is heavily dependent on the quality of the input EEG data. Artifacts and noise, which are common in clinical EEG recordings, can significantly impact the model's accuracy. While our model showed robustness to noise, further work is needed to enhance its resilience, especially in the context of real-world data with varying quality. Second, while the model generalizes well across the datasets used in this study, its performance in diverse clinical settings with different patient populations remains to be seen. The external validity of the model may be limited by the characteristics of the datasets used for training and testing. Additionally, the model's interpretability is a critical area for future research. While the hybrid model has shown promising results, the lack of transparency in its decision-making process poses challenges for clinical adoption. Efforts to interpret the model's attention weights and understand its decision basis are essential for building trust and acceptance in clinical practice. Lastly, the computational complexity of the model should be considered. The GATv2 and transformer architectures, while powerful, may demand significant computational resources, which could limit their applicability in resource-constrained settings.

4. Conclusion

In conclusion, this study demonstrates the efficacy of the GATv2 and transformer integrated model in classifying EEG signals. The model's superior performance, robustness to noise, and ability to generalize across datasets position it as a potent tool for both research and clinical applications. As we move forward, the continued development and refinement of this model hold great promise for enhancing our ability to diagnose and understand neurological disorders, ultimately contributing to improved patient outcomes. The implications of this research extend beyond EEG signal classification, offering a new

perspective on the analysis of graph-structured data across various domains.

Looking ahead, there is a clear path for further research and development. Optimization of the model's parameters, exploration of different architectures, and investigation into the model's behavior across a wider range of EEG datasets can all contribute to enhancing its performance. Additionally, the interpretability of the model's decisions is an area that warrants deeper exploration, as understanding the basis for the model's classifications can provide valuable insights into the underlying neurological processes. Finally, by further optimizing the model architecture to reduce computational complexity and improve processing speed, we can make it more practical for widespread clinical use. The implications of this research extend beyond EEG signal classification. The innovative combination of graph-based neural network and self-attention mechanism offers a new perspective on the analysis of graph-structured data across various domains, such as analyzing brain connectivity in other neurological diseases or even in nonmedical fields where graph-based data analysis is required. This paves the way for future research to explore the application of similar integrated models in different areas, ultimately contributing to improved patient outcomes and broader scientific advancements.

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Jian Lian and Zhenhua Xie contributed equally to this work.

Disclosures

The authors declare no conflict of interest. The founding sponsors had no role in the design of the study, in the collection, analysis, or interpretation of data; in the writing of the paper, or in the decision to publish the results.

Data Availability Statement

The CHB-MIT dataset used in this study can be downloaded from <https://physionet.org/static/published-projects/chbmit/chb-mit-scalp-eeg-database-1.0.0.zip>.

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